THE SMOOTH LOCUS OF A REDUCED LOCALLY ALGEBRAIC SCHEME OVER A PERFECT FIELD IS A DENSE OPEN AND EQUALS THE REGULAR LOCUS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author gives two independent arguments for density and labels them (1) and (2) in the source; both are reproduced as written.

Editorial note. The counted spans carry 6 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-dense-smooth-open-variety-over-perfect-field is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Perfect field and separability of residue field extensions. Algebra, Definition algebra-definition-perfect and Lemma algebra-lemma-separable-smooth, invoked at proof lines 4741-4744. Supplies the equality between the smooth locus and the regular locus.
- The smooth locus of a morphism of schemes is open. Morphisms, Definition morphisms-definition-smooth, invoked at proof lines 4744-4746. Supplies openness.
- The local ring of a reduced scheme at a generic point of a component is a field. Algebra, Lemma algebra-lemma-minimal-prime-reduced-ring, invoked at proof lines 4748-4750. Supplies the first density argument.
- Reducedness over a perfect field implies geometric reducedness, and the dense smooth open criterion. Same chapter, Lemmas lemma-perfect-reduced and lemma-geometrically-reduced-dense-smooth-open, invoked at proof lines 4751-4753. Supplies the second density argument.

Source reference. Original definition 030Y: sources/stacks-chow/tags.txt, source lines 3965–3965. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00EU: sources/stacks-chow/tags.txt, source lines 548–548. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 00TV: sources/stacks-chow/tags.txt, source lines 1089–1089. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 056V: sources/stacks-varieties/pinned-original-source.tex, source lines 4694–4694. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 020I: sources/stacks-varieties/pinned-original-source.tex, source lines 401–401. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 01V5: sources/stacks-chow/tags.txt, source lines 2434–2434. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a perfect field. Let X be a locally algebraic reduced k-scheme, for example a variety over k. Then we have*

$$\{x \in X \mid X \to \mathrm{Spec}(k) \text{ is smooth at } x\} = \{x \in X \mid \mathcal{O}_{X,x} \text{ is regular}\}$$

and this is a dense open subscheme of X.

Proof. The equality of the two sets follows immediately from Algebra, Lemma 00TV and the definitions (see Algebra, Definition 030Y for the definition of a perfect field). The set is open because the set of points where a morphism of schemes is smooth is open, see Morphisms, Definition 01V5. Finally, we give two arguments to see that it is dense: (1) The generic points of X are in the set as the local rings at generic points are fields (Algebra, Lemma 00EU) hence regular. (2) We use that X is geometrically reduced by Lemma 020I and hence Lemma 056V applies. □